

Generative Modeling for Molecules and Materials

Summer bootcamp about GenAI for chemistry and materials science

Course premise

This course assumes participants already know the high-level story of diffusion models, score models, flow matching, and normalizing flows. The purpose is not to reteach the basics. Instead, we lay out these basics in only the first two weeks, then move quickly into the current frontier: scalable flow matching, stochastic interpolants, couplings, one-step and few-step generation, discrete and hybrid models, guidance and alignment, Schrodinger bridges, and molecular/materials SOTA.

Each week has one compact theory target, a methods cluster, a chemistry/materials/biology application cluster, and a problem or coding assignment. The assignments are generally of the following styles: derive the identity behind the method, implement a tiny version, or critique why the method does or does not survive contact with scientific data.

Main references and section map

Text	How we use it	Link
MIT notes	Use Sections 1.3, 2, 3, 4, 5, 6, 7 and Appendices A to C. The notes are the quick review spine for ODE/SDE-based generation, flow matching, score matching, guidance, large-scale generators, and CTMC diffusion.	link
Stochastic Interpolants	Use Sections 2, 3, 5, and 6 as the conceptual bridge between flows, diffusions, deterministic samplers, stochastic samplers, and bridge-like methods.	link
Foundations of Schrodinger Bridges	Use Sections 1, 2, 3, 4.7, 6, 7, and 8 for OT, stochastic control, path-space KL, discrete bridges, and scientific applications such as data translation and Boltzmann sampling.	link
DMol	Use Chapters 8 to 10 and 15 to 16 for graphs, equivariance, normalizing flows, and modern molecular neural networks. Use application chapters as needed.	link

Weekly breakdown

Week 1: Probability transport, flow maps, and scientific state spaces

Theory target: Flow maps, velocity fields, continuity equation, log-density evolution, and the symmetries of molecular and crystalline state spaces.

Methods: Neural ODEs, FFJORD, flow matching as simulation-free CNF training, stochastic-interpolant normalizing flows.

Scientific connection: Boltzmann Generators, Equivariant Flows, early molecular normalizing-flow examples.

Week 2: Diffusion, score models, and modern design space

Theory target: Compressed review of DDPMs, score SDEs, reverse-time SDEs, probability-flow ODEs, denoising score matching, and sampler design.

SOTA methods: DDPM, score SDE, EDM, improved EDM, parameterization and preconditioning choices.

Scientific connection: EDM for 3D molecules, GeoDiff, TargetDiff, molecular conformers and ligand generation.

Week 3: Flow matching, rectified flows, and scalable transformer generators

Theory target: Conditional flow matching, marginal vector fields, straightness, path curvature, and why flow matching scales well.

Methods: Flow Matching, Rectified Flow, Flow Matching Guide and Code, Stable Diffusion 3, Movie Gen.

Scientific connection: FoldFlow, FlowMM, crystal and protein flow matching.

Assignment: Prove CFM unbiasedness. Train a toy CFM model and compare independent versus OT pairings.

Week 4: Stochastic interpolants as the unifying language

Theory target: Interpolants, velocity objectives, score objectives, deterministic and stochastic samplers, and recovery of diffusion and flow-matching limits.

Methods: Stochastic Interpolants, data-dependent couplings, Scalable Interpolant Transformers.

Scientific connection: OMatG, MatterGen, CrystalFlow, stochastic-interpolant crystal generation.

Week 5: Couplings, optimal transport, and data-dependent bases

Theory target: Why couplings determine path crossing, stiffness, curvature, and model capacity. Entropic OT and minibatch Sinkhorn as practical tools.

Methods: OT-CFM, rectification and reflow, data-dependent stochastic interpolants, learned bases.

Scientific connection: Low-fidelity to high-fidelity ensembles, coarse-to-fine simulation, composition-conditioned crystal generation, Boltzmann sampling.

Week 6: One-step and few-step generation

Theory target: Finite-time flow maps, semigroup consistency, average velocity, distillation versus training from scratch, and the cost of fast sampling.

Methods: Consistency Models, Progressive Distillation, Align Your Flow, MeanFlow, Inductive Moment Matching, flow-map language models, Flow Map Reward Guidance.

Scientific connection: Fast screening loops, active learning, docking loops, MLIP or DFT post-relaxation, test-time guided design.

Week 7: Discrete, categorical, and hybrid continuous-discrete generation

Theory target: CTMC diffusion, score ratios, probability-simplex geometry, masked generation, and continuous embedding flows.

Methods: SEDD, Mercury, Discrete Flow Maps, ELF, masked diffusion, MAR with diffusion loss.

Scientific connection: DiGress, SELFIES or SMILES generation, protein sequences, atom and bond types, crystal compositions and site identities.

Week 8: Guidance, alignment, and inverse design

Theory target: Guidance as reward tilting, score correction, stochastic control, surrogate optimization, and Pareto design.

Methods: Classifier guidance, classifier-free guidance, Diffusion-DPO, preference-guided diffusion, Flow Map Reward Guidance, ChemGuide.

Scientific connection: Stability, docking, binding affinity, formation energy, band gap, elastic constants, synthesizability, and multi-objective design.

Week 9: Schrodinger bridges, stochastic control, and ensemble translation

Theory target: Static and dynamic SB, path-space KL, stochastic optimal control, Girsanov, bridge matching, and endpoint-constrained dynamics.

Methods: Diffusion Schrodinger Bridge, simulation-free SB via score and flow matching, bridge matching.

Scientific connection: Single-cell state dynamics, Boltzmann distributions, unrelaxed to relaxed structures, low-fidelity to high-fidelity simulation.

Week 10: Molecular, protein, and biomolecular generative SOTA

Theory target: Coordinate, torsion, and frame state spaces. $E(3)$ and $SE(3)$ equivariance. Pocket conditioning and sequence-structure co-design.

Methods: EDM for 3D molecules, GeoDiff, TargetDiff, FrameDiff, FoldFlow, RFDiffusion, Chroma, AlphaFold 3.

Scientific connection: Small molecules, conformers, ligands, protein binders, scaffolds, biomolecular complexes.

Week 11: Crystalline materials generative SOTA

Theory target: Fractional coordinates, periodicity, lattice generation, space-group structure, atom types, and stability metrics.

Methods: CDVAE, DiffCSP, MatterGen, FlowMM, OMatG, CrystalFlow.

Scientific connection: CSP, de novo inorganic generation, property-conditioned materials design, DFT or MLIP validation.

Week 12: Failure modes and evaluation

Theory target: OOD generalization, extrapolation, surrogate hacking, benchmark leakage, relaxation artifacts, uncertainty, and negative results.

Methods: Critical synthesis of MatterGen, FlowMM, OMatG, CrystalFlow, RFdiffusion, Chroma, AlphaFold 3, and one participant-selected recent paper.

Scientific connection: What makes a generated molecule or material useful rather than merely valid.